## C PROOFS

*Proof.* **Theorem 1**
First we introduce the notation $C_1 = S$ and $C_2 = X \backslash S$ and note that $C_1$ and $C_2$ form a partition of $X$, i.e. $X = C_1 \cup C_2$ with $C_1 \cap C_2 = \emptyset$. We note $C'_j$ the set of indices of $C_j$, meaning that $i \in C'_j \Leftrightarrow x_i \in C_j$. We then rewrite $X_{S,\vec{v},c}$ as being the union of the set of points in $C_1$ and the points in $C_2$ translated by $c$ in direction $\vec{v}$:

$$X_{S,\vec{v},c} = \{x : x \in C_1\} \cup \{x + c\vec{v} : x \in C_2\}$$

It is standard practice to center the two sets of representations being compared before using a representation similarity measures. $X$ is centered by hypothesis but $X_{S,\vec{v},c}$ is not. We first note that the mean of $X_{S,\vec{v},c}$ across all representations $\overline{X}_{S,\vec{v},c}$ is the vector:

$$\begin{aligned}
\overline{X}_{S,\vec{v},c} &= \frac{1}{n} \sum_{x \in X_{S,\vec{v},c}} x \\
&= \frac{1}{n} \left( \sum_{i \in C'_1} x_i + \sum_{i \in C'_2} x_i + c\vec{v} \right) && \text{by definition of } X_{S,\vec{v},c} \\
&= \frac{1}{n} \left( \sum_{i \in C'_1 \cup C'_2} x_i + \sum_{i \in C'_2} c\vec{v} \right) \\
&= \frac{1}{n} \sum_{x \in X} x + \frac{1}{n} \sum_{i \in C'_2} c\vec{v} && \text{because } C_1, C_2 \text{ form a partition of } X \\
&= \frac{|C_2| c\vec{v}}{n} && \text{because } X \text{ is centered by hypothesis}
\end{aligned}$$

From now on we note $Y$ as being the centered set of representations $X_{S,\vec{v},c}$ where we subtracted the mean $\overline{X}_{S,\vec{v},c}$ (here we used the fact that $|C_1| + |C_2| = n$):

$$Y = \{x - \frac{|C_2| c\vec{v}}{n} : x \in C_1\} \cup \{x + \frac{|C_1| c\vec{v}}{n} : x \in C_2\}$$

Now that we have workable expressions for $X$ and $X_{S,\vec{v},c}$ we focus on the computation of linear CKA which relies on the computation of three HSIC values: between $X$ and itself, between $Y$ and itself and between $X$ and $Y$:

$$\text{CKA}_{lin}(X, Y) = \frac{\text{HSIC}_{lin}(X, Y)}{\sqrt{\text{HSIC}_{lin}(X, X)\text{HSIC}_{lin}(Y, Y)}} \tag{5}$$

We also remind the reader that linear HSIC takes the form:

$$\text{HSIC}_{lin}(X, Y) = \frac{1}{(n-1)^2} tr(XX^\top YY^\top) = \frac{1}{(n-1)^2} \sum_{i=1}^{n} \sum_{j=1}^{n} \langle x_i, x_j \rangle \langle y_j, y_i \rangle \tag{6}$$

We can split the terms of the two sums into three distinct categories and compute the values of the inner products independently in terms of $x_i, x_j$ and $c$ for the three HSIC terms:

1. $i \in C_1'$ and $j \in C_1'$ (i.e. $x_i \in C_1$ and $x_j \in C_1$):

$$
\begin{aligned}
\langle x_i, x_j\rangle^2 &= \langle x_i, x_j\rangle^2 \\
\langle x_i, x_j\rangle\langle y_i, y_j\rangle &= \langle x_i, x_j\rangle\langle x_i - \frac{c|C_2'|}{n}\vec{v}, x_j - \frac{c|C_2'|}{n}\vec{v}\rangle \\
&= \langle x_i, x_j\rangle\left[\langle x_i, x_j\rangle - \frac{c|C_2'|}{n}\langle x_i, \vec{v}\rangle - \frac{c|C_2'|}{n}\langle \vec{v}, x_j\rangle + \left(-\frac{c|C_2'|}{n}\right)^2\langle \vec{v}, \vec{v}\rangle\right] \\
&= \langle x_i, x_j\rangle\left[\langle x_i, x_j\rangle - \frac{c|C_2'|}{n}\langle x_i, \vec{v}\rangle - \frac{c|C_2'|}{n}\langle \vec{v}, x_j\rangle + \frac{c^2|C_2'|^2}{n^2}\right] \\
&= \langle x_i, x_j\rangle^2 - \frac{c|C_2'|}{n}\langle x_i, \vec{v}\rangle\langle x_i, x_j\rangle - \frac{c|C_2'|}{n}\langle \vec{v}, x_j\rangle\langle x_i, x_j\rangle + \frac{c^2|C_2'|^2}{n^2}\langle x_i, x_j\rangle \\
&= \mathcal{O}(c) + \frac{c^2|C_2'|^2}{n^2}\langle x_i, x_j\rangle \\
\langle y_i, y_j\rangle^2 &= \langle x_i - \frac{c|C_2'|}{n}\vec{v}, x_j - \frac{c|C_2'|}{n}\vec{v}\rangle^2 \\
&= \left[\langle x_i, x_j\rangle - \frac{c|C_2'|}{n}\langle x_i, \vec{v}\rangle - \frac{c|C_2'|}{n}\langle \vec{v}, x_j\rangle + \frac{c^2|C_2'|^2}{n^2}\right]^2 \\
&= \mathcal{O}(c^3) + \frac{c^4|C_2'|^4}{n^4}
\end{aligned}
$$

2. $i \in C_1'$ and $j \in C_2'$ (i.e. $x_i \in C_1$ and $x_j \in C_2$):

$$
\begin{aligned}
\langle x_i, x_j\rangle^2 &= \langle x_i, x_j\rangle^2 \\
\langle x_i, x_j\rangle\langle y_i, y_j\rangle &= \langle x_i, x_j\rangle\langle x_i - \frac{c|C_2'|}{n}\vec{v}, x_j + \frac{c|C_1'|}{n}\vec{v}\rangle \\
&= \langle x_i, x_j\rangle\left[\langle x_i, x_j\rangle + \frac{c|C_1'|}{n}\langle x_i, \vec{v}\rangle - \frac{c|C_2'|}{n}\langle \vec{v}, x_j\rangle - \frac{c^2|C_1'||C_2'|}{n^2}\langle \vec{v}, \vec{v}\rangle\right] \\
&= \langle x_i, x_j\rangle\left[\langle x_i, x_j\rangle + \frac{c|C_1'|}{n}\langle x_i, \vec{v}\rangle - \frac{c|C_2'|}{n}\langle \vec{v}, x_j\rangle - \frac{c^2|C_1'||C_2'|}{n^2}\right] \\
&= \langle x_i, x_j\rangle^2 + \frac{c|C_1'|}{n}\langle x_i, x_j\rangle\langle x_i, \vec{v}\rangle - \frac{c|C_2'|}{n}\langle x_i, x_j\rangle\langle \vec{v}, x_j\rangle - \frac{c^2|C_1'||C_2'|}{n^2}\langle x_i, x_j\rangle \\
&= \mathcal{O}(c) - \frac{c^2|C_1'||C_2'|}{n^2}\langle x_i, x_j\rangle \\
\langle y_i, y_j\rangle^2 &= \langle x_i - \frac{c|C_2'|}{n}\vec{v}, x_j + \frac{c|C_1'|}{n}\vec{v}\rangle^2 \\
&= \left[\langle x_i, x_j\rangle + \frac{c|C_1'|}{n}\langle x_i, \vec{v}\rangle - \frac{c|C_2'|}{n}\langle \vec{v}, x_j\rangle - \frac{c^2|C_1'||C_2'|^2}{n^2}\right]^2 \\
&= \mathcal{O}(c^3) + \frac{c^4|C_1'|^2|C_2'|^2}{n^4}
\end{aligned}
$$

3. $i \in C_2'$ and $j \in C_2'$ (i.e. $x_i \in C_2$ and $x_j \in C_2$):

$$
\begin{aligned}
\langle x_i, x_j \rangle^2 &= \langle x_i, x_j \rangle^2 \\
\langle x_i, x_j \rangle \langle y_i, y_j \rangle &= \langle x_i, x_j \rangle \langle x_i + \frac{c|C_1'|}{n} \vec{v}, x_j + \frac{c|C_1'|}{n} \vec{v} \rangle \\
&= \langle x_i, x_j \rangle \left[ \langle x_i, x_j \rangle + \frac{2c|C_1'|}{n} \langle x_i, \vec{v} \rangle + \left( + \frac{c|C_1'|}{n} \right)^2 \langle \vec{v}, \vec{v} \rangle \right] \\
&= \langle x_i, x_j \rangle \left[ \langle x_i, x_j \rangle + \frac{2c|C_1'|}{n} \langle x_i, \vec{v} \rangle + \frac{c^2|C_1'|^2}{n^2} \right] \\
&= \langle x_i, x_j \rangle^2 + \frac{2c|C_1'|}{n} \langle x_i, x_j \rangle \langle x_i, \vec{v} \rangle + \frac{c^2|C_1'|^2}{n^2} \langle x_i, x_j \rangle \\
&= \mathcal{O}(c) + \frac{c^2|C_1'|^2}{n^2} \langle x_i, x_j \rangle \\
\langle y_i, y_j \rangle^2 &= \langle x_i + \frac{c|C_1'|}{n} \vec{v}, x_j + \frac{c|C_1'|}{n} \vec{v} \rangle^2 \\
&= \left[ \langle x_i, x_j \rangle + \frac{2c|C_1'|}{n} \langle x_i, \vec{v} \rangle + \frac{c^2|C_1'|^2}{n^2} \right]^2 \\
&= \mathcal{O}(c^3) + \frac{c^4|C_1'|^4}{n^4}
\end{aligned}
$$

When we take $\lim_{c \to \infty} CKA_{lin}(X, Y) = \lim_{c \to \infty} \frac{\text{HSIC}_{lin}(X,Y)}{\sqrt{\text{HSIC}_{lin}(X,X)\text{HSIC}_{lin}(Y,Y)}}$, it is easy to see that the terms with the highest powers of $c$ will dominate the expression. At the numerator that is $c^2$ and at the denominator that is $c^4$ inside the square root. To convince oneself of this it suffices to divide by $c^2$ at the numerator and at the denominator, all terms except the higher power ones will then tend to 0 as $c$ tends to infinity, so at the limit we have: